Compiler Construction

Assignment # 01 **Marks: 20** **Due Date: 16.03.26**

Q1: Explain Lexical Analyzer Generator tool, Structure of Lex/Flex program, Sample Lex programs. Also explain other Compiler Construction Tools.

Your task is to write a program that reads an input text file, and constructs a list of tokens in

that file. There will be two columns. One column will display list of tokens and other will display Lexemes. Your program may be written in C, C++, Java or any other programming language.

Assuming that the input file contains the following code string:

```
void main ()

{

int sum = 0;

for(int j=0; j < 10; j=j+1)

{

sum = sum + j + 10.43 + 344 + 45.34E-4 + 43.0 + 7.34;

}

}
```

Q2: Give state diagrams of NFA's with the specified number of states for each of the following

Languages over the alphabet {0, 1}.

1. L1 = {w|w ends with 01}, with 3 states

2. L2 = {w|w contains an even number of 0's, or even no of 1's}, with 3 states

Q3. Describe the role of lexical Analysis phase and syntax Analysis phase of compiler construction.

Q4. Remove Recursion from Grammar and identify the First and Follow sets from the grammar. Construct the Predictive Descent parser table (M-Table).

$S \rightarrow aS \mid bS \mid cS \mid dS \mid e \mid f \mid g \mid rAg$

$A \rightarrow h \mid u \mid \Lambda \mid Sab \mid SBb \mid SAh$

$B \rightarrow bBg \mid bbB \mid Bfc \mid \Lambda$